



**Year 11 Mathematics Advanced
Assessment Task 1
Term 2, Week 3, 2024**

Setter: BRA

Name: _____

Teacher: _____

- Answer Questions in the space provided.
- Answer multiple choice questions on the question paper by circling the correct alternative.
- Show all relevant mathematical reasoning and/or calculations.
- Use blue or black PEN only.
- NESA approved calculators are permitted in this task.
- Your own Formula Sheet can be used with this task.
- Marks may be deducted for insufficient or illegible work.
- Extra writing space is provided at the back of this booklet.
If you use this space, clearly indicate which question you are answering.
- Total possible marks: **32**
- Time allowed: **45 minutes plus 2 minutes Reading Time**

Wednesday , 15th May 2024

Q1	Q2	Q3	Q4	Q5	Q6	Q7	TOTAL
/6	/6	/4	/4	/4	/4	/4	/32

Question 1 (6 marks)**Marks**

(a) Circle your correct alternative.

1

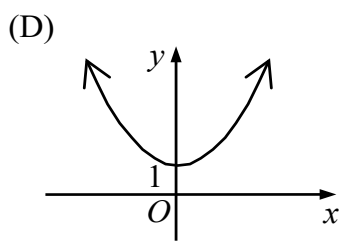
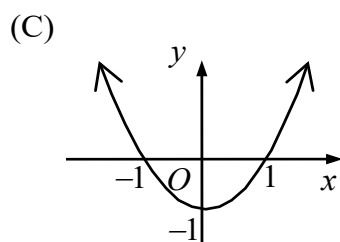
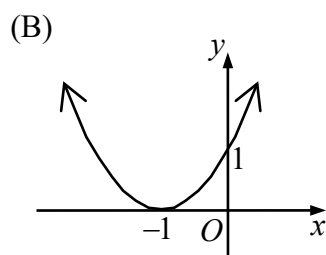
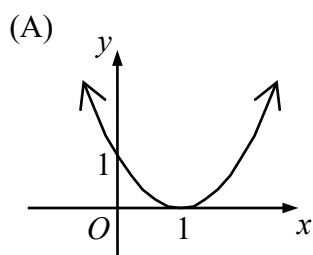
If $ab = 12$, $bc = 20$ and $ac = 15$, then a possible value for abc is:

(A) -600 (B) 60 (C) 360 (D) 600

(b) Circle your correct alternative.

1

Which graph drawn below best represents $y = (x + 1)^2$?



Question 1 Continued

- (c) Given $M = \frac{m}{\sqrt{1 - \frac{v^2}{c^2}}}$, find M if $m = 550$, $v = 2.75 \times 10^8$ and $c = 3 \times 10^8$.

2

Give your answer correct to five significant figures.

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- (d) Simplify completely $\frac{4a^2b - 8ab^2}{4ab^2 - 8a^2b}$.

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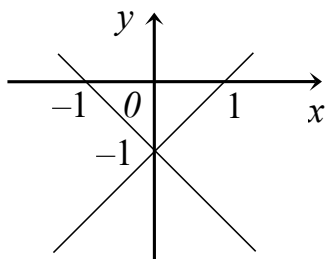
End of Question 1

Question 2 (6 marks)**Marks**

- (a) Circle your correct alternative.

1

The equation $x^2 - (y+1)^2 = 0$ represents a relation between x and y in the xy -plane. Its sketch is shown below.



The relation can be classified as:

- (A) One-to-one
- (B) One-to-many
- (C) Many-to-one
- (D) Many-to-many

- (b) Circle your correct alternative.

1

Consider the following two statements:

- I. Two irrational zeroes of a quadratic function cannot be equal.
- II. If α and β are distinct zeroes of the quadratic function $f(x) = ax^2 + bx + c$, where a , b and c are real constants, then $a\alpha + b + a\beta = 0$.

Which of the above statements are true?

- (A) Neither I nor II
- (B) I only
- (C) II only
- (D) Both I and II

Question 2 Continued

(c) Solve $\frac{x-5}{4} = \frac{6}{x}$.

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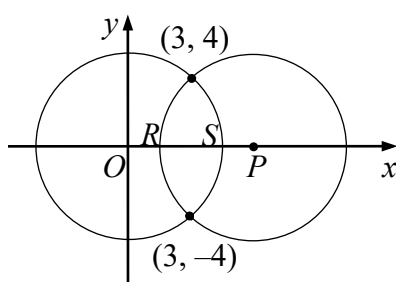
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(d)



Not to Scale

2

In the diagram above, O and P are centres of circles with equal radii, which intersect at $(3, 4)$ and $(3, -4)$. The circles cut the x -axis at R and S as shown.

Find the value of the length RS . Explain your reasoning carefully.

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End of Question 2

Question 3 (4 marks)**Marks**

(a) Sketch a function $f(x)$ that possesses all of the following properties:

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1. $f(x)$ is an odd function.
2. $f(x)$ has horizontal asymptotes $y = \pm 2$.
3. $f(x)$ has vertical asymptotes $x = \pm 2$.
4. $f(x)$ has domain $x < -2$ together with $x > 2$ as well as $x = 0$.

(b) Write $\frac{\sqrt{2}+1}{\sqrt{2}-1}$ in the form of $a + b\sqrt{x}$, where a, b and x are integers.

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End of Question 3

Question 4 (4 marks)**Marks**

A function is defined by the formula $f(x) = \frac{\sqrt{-x}}{\sqrt{x+1}}$.

- (a) Determine the domain of the function.

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- (b) Determine the range of the function.

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- (c) Given f is a one-to-one function, what is the relationship between x_1 and x_2 , if $f(x_1) = f(x_2)$? Explain your reasoning carefully.
You may assume x_1 and x_2 are in the domain of f .

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End of Question 4

Question 5 (4 marks)

- (a) If $f(x) = \sqrt{x}$ sketch the graph of $y = f(x-1)$ showing intercepts with the coordinate axes. **1**

- (b) $P(1, 2)$ is a point on the line $3x + y - 5 = 0$. Find the coordinates of the points on this line, so that their distance from P is $\sqrt{10}$ units. **3**

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End of Question 5

Question 6 (4 marks)

- (a) Simplify completely $\sqrt{54x} \div \sqrt{6}$

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- (b) Factorise fully $a^2 - 2ab + b^2 + 3a - 3b + 2$.

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End of Question 6

Question 7 (4 marks)

- (a) How many integers are between $\sqrt{50}$ and $\sqrt{500}$?

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- (b) By definition, a positive integer that can be written as the square of a positive integer is called a square number or a perfect square. For example, 9 is a perfect square since $9 = 3^2$. 3

For x a positive integer, how many positive integers are there such that both x and $x + 99$ are perfect squares? Explain carefully your reasoning.

Hint: Let $x = m^2$ and $x + 99 = n^2$, where m and n are positive integers.

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

End of Assessment Task

Extra writing space

If you are using this space indicate clearly which question(s) you are answering & write ‘see extra writing space’ on the original question space provided for that question.

[illegible]

[illegible]



**Year 11 Mathematics Advanced
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/6	/6	/4	/4	/4	/4	/4	/32

Question 1 (6 marks)

Marks

(a) Circle your correct alternative.

1

If $ab = 12$, $bc = 20$ and $ac = 15$, then a possible value for abc is:

(A) - 600

(B) 60

(C) 360

(D) 600

$$(ab)(ac) = 12 \times 15$$

$$\Rightarrow a^2 bc = 180 \quad \& \quad bc = 20$$

$$\Rightarrow 20a^2 = 180$$

$$\Rightarrow a^2 = 9$$

$$\Rightarrow a = \pm 3$$

$$\therefore abc = a \times (bc)$$

$$= \pm 3 \times 20$$

$$= \pm 60$$

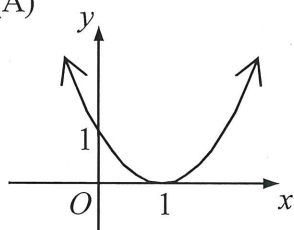
$\therefore (B)$

(b) Circle your correct alternative.

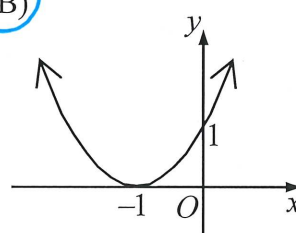
1

Which graph drawn below best represents $y = (x + 1)^2$?

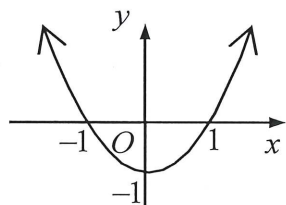
(A)



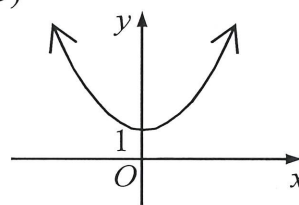
(B)



(C)



(D)



$y = (x+1)^2$ is a translation of $y = x^2$ one unit to the LEFT. $\therefore (B)$.

Question 1 Continued

- (c) Given $M = \frac{m}{\sqrt{1 - \frac{v^2}{c^2}}}$, find M if $m = 550$, $v = 2.75 \times 10^8$ and $c = 3 \times 10^8$.

2

Give your answer correct to five significant figures.

$$M = \frac{550}{\sqrt{1 - \left(\frac{2.75 \times 10^8}{3 \times 10^8}\right)^2}}$$

N.B. $\frac{2.75}{3} = \frac{11}{12}$ & so equivalent calculation is

$$= \frac{550}{\sqrt{1 - \left(\frac{2.75}{3}\right)^2}}$$

$$= \frac{550}{\sqrt{1 - \left(\frac{11}{12}\right)^2}}$$

$$= 1376.195133... \text{ (Calculator)}$$

$$= 1376.2 \text{ (5 s.f.)}$$

Correct bald
answer: 2 MARKS

- (d) Simplify completely $\frac{4a^2b - 8ab^2}{4ab^2 - 8a^2b}$.

$$= \frac{4ab(a - 2b)}{4ab(b - 2a)} \quad \checkmark \text{ 1 mark}$$

$$= \frac{a - 2b}{b - 2a} \quad \checkmark \text{ 1 mark}$$

BALD ANSWER: 1 mark only.

✓ for correct calculation
✓ for correct # of sig. figures. 2

End of Question 1

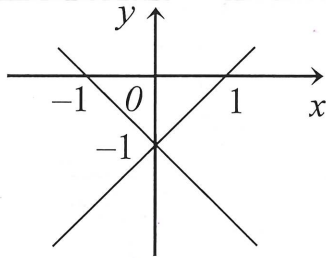
Question 2 (6 marks)

Marks

(a) Circle your correct alternative.

1

The equation $x^2 - (y+1)^2 = 0$ represents a relation between x and y in the xy -plane. Its sketch is shown below.



The relation can be classified as:

- (A) One-to-one
- (B) One-to-many
- (C) Many-to-one
- ☒ (D) Many-to-many

The relation is NOT a function and so (A) & (C) are immediately eliminated. As the relation fails the horizontal line test as well, then it is classified as many-to-many. So (D)

(b) I. is false. Counterexample:

$$f(x) = x^2 - 2\sqrt{2}x + 2, \text{ i.e., } f(x) = (x - \sqrt{2})^2 \text{ whose zeroes are } \sqrt{2}, \sqrt{2} \text{ or equivalently } \sqrt{2}(2).$$

It is true. As α and β are zeroes of $f(x)$, then: -

(b) Circle your correct alternative.

$$a\alpha^2 + b\alpha + c = 0 \quad \& \quad 1$$

$$a\beta^2 + b\beta + c = 0$$

Consider the following two statements:

- I. Two irrational zeroes of a quadratic function cannot be equal.
- II. If α and β are distinct zeroes of the quadratic function $f(x) = ax^2 + bx + c$, where a, b and c are real constants, then $a\alpha + b + a\beta = 0$.

Which of the above statements are true?

- (A) Neither I nor II
- (B) I only
- ☒ (C) II only
- (D) Both I and II

Subtracting the above equations yields $a(\alpha^2 - \beta^2) + b(\alpha - \beta) = 0$
 $\Rightarrow a(\alpha - \beta)(\alpha + \beta) + b(\alpha - \beta) = 0$
 $\Rightarrow a(\alpha + \beta) + b = 0$ as $\alpha \neq \beta$
 $\Rightarrow a\alpha + b + a\beta = 0$, as required.
 $\therefore (C)$

Question 2 Continued

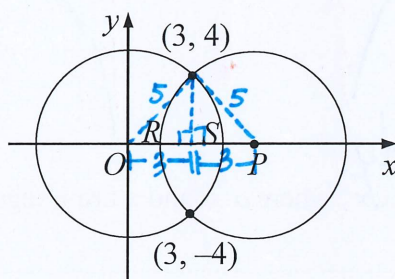
(c) Solve $\frac{x-5}{4} = \frac{6}{x}$.

2

$$\begin{aligned} \Rightarrow x(x-5) &= 24 \\ \Leftrightarrow x^2 - 5x - 24 &= 0 \quad \checkmark \text{ correct quadratic equation} \\ \Leftrightarrow (x-8)(x+3) &= 0 \\ \Leftrightarrow x &= -3 \text{ or } 8 \quad \checkmark \text{ correct answers.} \end{aligned}$$

(d)

2



Not to Scale

In the diagram above, O and P are centres of circles with equal radii, which intersect at $(3, 4)$ and $(3, -4)$. The circles cut the x axis at R and S as shown.

Find the value of the length RS . Explain your reasoning carefully.

Radius of both circles found by a simple application of Pythagoras' theorem or the distance formula applied to $O(0,0)$ & $(3,4)$. Radius = 5 units.
 From the same right-angled triangle constructions, it is evident that $OP = 3 + 3$ or 6 units. $\therefore OR = OP - RP = 6 - 5$ or 1 unit. Finally, $RS = OS - OR$
 $\checkmark$ Correct answer $= \text{Radius} - OR$
 $\checkmark$ Valid reasoning. $= 5 - 1$ or 4 units.

End of Question 2

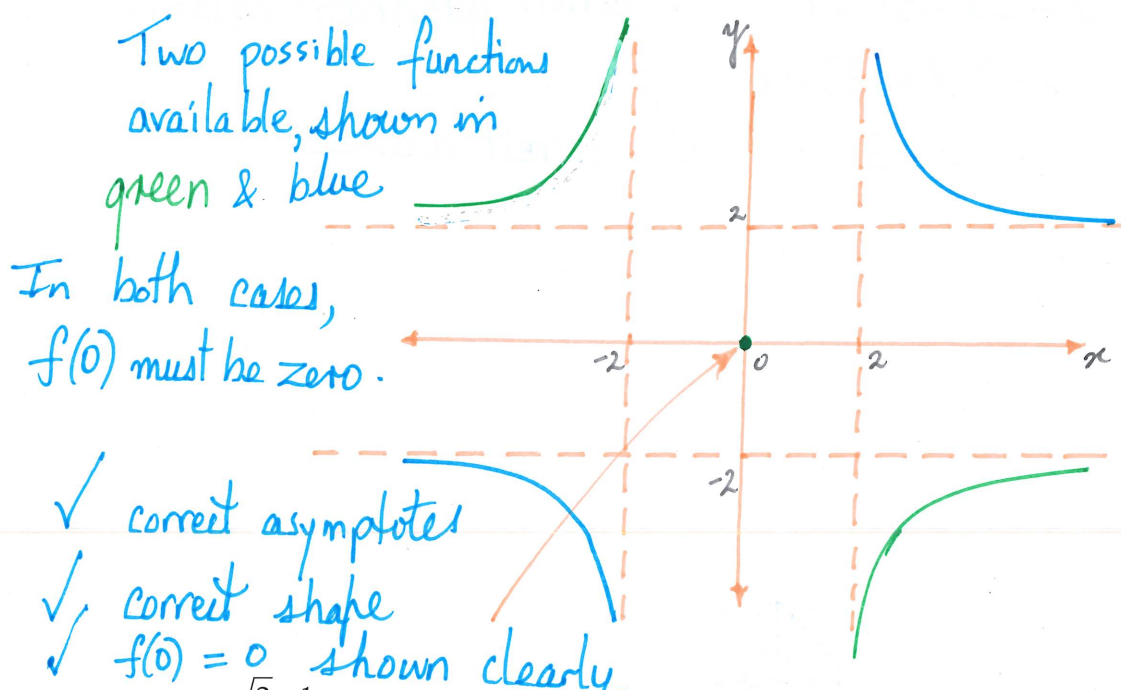
Question 3 (4 marks)

Marks

(a) Sketch a function $y = f(x)$ that possesses all of the following properties:

3

1. $f(x)$ is an odd function.
2. $f(x)$ has horizontal asymptotes $y = \pm 2$.
3. $f(x)$ has vertical asymptotes $x = \pm 2$.
4. $f(x)$ has domain $x < -2$ together with $x > 2$ as well as $x = 0$.



(b) Write $\frac{\sqrt{2}+1}{\sqrt{2}-1}$ in the form of $a + b\sqrt{x}$, where a , b and x are integers.

1

$$\begin{aligned} \frac{\sqrt{2}+1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1} &= \frac{(\sqrt{2}+1)^2}{1} \\ &= (\sqrt{2})^2 + 2 \times 1 \times \sqrt{2} + 1^2 \\ &= \underline{\underline{3 + 2\sqrt{2}}} \quad \checkmark \text{ Correct answer.} \end{aligned}$$

End of Question 3

Question 4 (4 marks)

Marks

A function is defined by the formula $f(x) = \frac{\sqrt{-x}}{\sqrt{x+1}}$.

- (a) Determine the domain of the function.

2

Require $-x \geq 0$ & $x+1 > 0$ ✓ Correct inequalities
 $\Rightarrow x \leq 0$ & $x > -1$ } ✓ Correct simplification.
 i.e. $-1 < x \leq 0$

- (b) Determine the range of the function.

1

Clearly $f(x)$ is non-negative. Indeed, when $x=0$, $f(0)=0$ and $f \rightarrow +\infty$ as $x \rightarrow -1^+$ (Calculator check is available.)
 So range is $y \geq 0$ ✓ Correct answer

- (c) Given f is a one-to-one function, what is the relationship between x_1 and x_2 , if $f(x_1) = f(x_2)$? Explain your reasoning carefully.

1

You may assume x_1 and x_2 are in the domain of f .

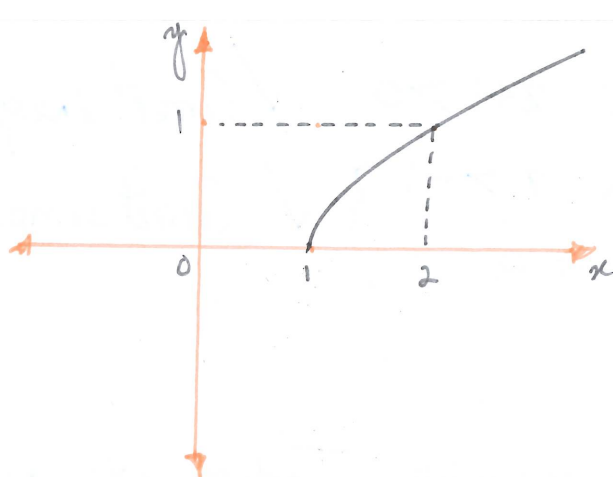
As f is a 1-1 function, then any horizontal line cuts the curve at most once. So if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

✓ For $x_1 = x_2$ with valid reasoning.

End of Question 4

Question 5 (4 marks)

- (a) If $f(x) = \sqrt{x}$ sketch the graph of $y = f(x-1)$ showing intercepts with the coordinate axes. 1



✓ correct shape with
x-intercept at $x = 1$.

- (b) $P(1, 2)$ is a point on the line $3x + y - 5 = 0$. Find the coordinates of the points on this line, so that their distance from P is $\sqrt{10}$ units. 3

✓
1 mark { Let $Q(x, y)$ be any other point on the line $3x + y - 5 = 0$.
Applying the distance formula: -
$$(x-1)^2 + (y-2)^2 = (\sqrt{10})^2 \quad [PQ^2 = 10]$$

$$\Rightarrow (x-1)^2 + (y-2)^2 = 10 \quad \dots (1)$$

1 mark { However, as $Q(x, y)$ lies on the line, then $y = 5 - 3x$
(from $3x + y - 5 = 0$). So (1) becomes: -
$$(x-1)^2 + (5-3x-2)^2 = 10$$

$$\Leftrightarrow (x-1)^2 + (3-3x)^2 = 10$$

$$\Leftrightarrow x^2 - 2x + 1 + 9 - 18x + 9x^2 = 10$$

$$\Leftrightarrow 10x^2 - 20x + 10 = 10$$

$$\Leftrightarrow 10x(x-2) = 0$$

$$\Rightarrow \begin{cases} x = 0 \text{ or } x = 2 \\ y = 5 \text{ or } y = -1 \text{ respectively.} \end{cases}$$

$$\therefore \text{required co-ordinates are } (0, 5) \text{ \& } (2, -1).$$

✓ correct answers.

End of Question 5

Question 6 (4 marks)

- (a) Simplify completely $\sqrt{54x} \div \sqrt{6}$

1

$$= \frac{\sqrt{54x}}{\sqrt{6}} \text{ or } \sqrt{\frac{54x}{6}}$$

$$= \sqrt{9x} \text{ or } \underline{3\sqrt{x}} \quad \checkmark \text{ for correct answer}$$

- (b) Factorise fully $a^2 - 2ab + b^2 + 3a - 3b + 2$.

3

$$= (a-b)^2 + 3a - 3b + 2$$

$$= (a-b)^2 + 3(a-b) + 2$$

Let $X = a-b$. Algebra is equivalent to

$$X^2 + 3X + 2$$

$$\Rightarrow (X+2)(X+1)$$

$$\Rightarrow (a-b+2)(a-b+1)$$

2 marks
or
equivalent
logic.

End of Question 6

Question 7 (4 marks)

- (a) How many integers are between $\sqrt{50}$ and $\sqrt{500}$?

1

Noting $\sqrt{8^2} = \sqrt{64}$ or 8 is the least & $\sqrt{22^2} = \sqrt{484}$ or 22 is the largest, we require an inclusive count from 8 through to 22. This is $22 - 8 + 1$ or 15. Hence, there are 15 integers between $\sqrt{50}$ & $\sqrt{500}$.
1 mark

- (b) By definition, a positive integer that can be written as the square of a positive integer is called a square number or a perfect square. For example, 9 is a perfect square since $9 = 3^2$.

3

For x a positive integer, how many positive integers are there such that both x and $x + 99$ are perfect squares? Explain carefully your reasoning.

Hint: Let $x = m^2$ and $x + 99 = n^2$, where m and n are positive integers.

Using the hint $n^2 - m^2 = (x + 99) - x$

i.e., $(n + m)(n - m) = 99$; $n > m$

1 mark or equivalent

Consider the factors of 99 which are 1, 3, 9, 11, 33 & 99.

Thus, possible combinations for n & m are:-

$$\begin{cases} n+m=99 \\ n-m=1 \end{cases}, \begin{cases} n+m=33 \\ n-m=3 \end{cases}, \begin{cases} n+m=11 \\ n-m=9 \end{cases}$$

Solving these systems of simultaneous equations gives:-

$n=50, m=49$; $n=18, m=15$; $n=10, m=1$ respectively.

Thus, x could be $\begin{cases} 49^2 \text{ or } 2401 \\ 15^2 \text{ or } 225 \\ 1^2 \text{ or } 1 \end{cases}$

Consequently, there are exactly three positive integers satisfying the criteria.

1 mark for systems of linear equations

1 mark for correct answer/conclusion following from argument

End of Assessment Task

N.B. No credit for "correct" bald answer.